

Regression Presentation

Robert Gulotty

11/6/2020

The Ordinary Least Squares Model

In many quantitative papers people display standard outputs from statistical software. Today we will cover three main ideas.

- ▶ How are the parameters of the model (β) calculated.
- ▶ How are standard errors calculated and summarized
- ▶ What diagnostics are reported.

Prestige Data

```
summary(Prestige[,c("prestige", "education", "income")])
```

##	prestige	education	income
##	Min. :14.80	Min. : 6.380	Min. : 611
##	1st Qu.:35.23	1st Qu.: 8.445	1st Qu.: 4106
##	Median :43.60	Median :10.540	Median : 5930
##	Mean :46.83	Mean :10.738	Mean : 6798
##	3rd Qu.:59.27	3rd Qu.:12.648	3rd Qu.: 8187
##	Max. :87.20	Max. :15.970	Max. :25879

Predicting Prestige with lm()

```
summary(lm(prestige~education+income, data=Prestige))
```

```
##
## Call:
## lm(formula = prestige ~ education + income, data = Prestige)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.4040  -5.3308   0.0154   4.9803  17.6889
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -6.8477787  3.2189771  -2.127   0.0359 *
## education    4.1374444  0.3489120  11.858 < 2e-16 ***
## income       0.0013612  0.0002242   6.071 2.36e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.81 on 99 degrees of freedom
## Multiple R-squared:  0.798, Adjusted R-squared:  0.7939
## F-statistic: 195.6 on 2 and 99 DF, p-value: < 2.2e-16
```

Visulazation of Prestige

```
## Loading required namespace: rgl
```

```
## Loading required namespace: mgcv
```

Output and Input

```
y<-Prestige$prestige
xvars<-as.matrix(Prestige[,c("education", "income")])
# Add vector of 1 to X matrix
xmat<-cbind(1,xvars)
```

$$\beta = (X'X)^{-1}X'y.$$

```
betas<-solve(t(xmat)%*%xmat)%*%t(xmat)%*%y
betas
```

```
##                [,1]
##            -6.847778720
## education   4.137444384
## income      0.001361166
```

Note Betas

```
summary(lm(prestige~education+income, data=Prestige))
```

```
##
## Call:
## lm(formula = prestige ~ education + income, data = Prestige)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.4040  -5.3308   0.0154   4.9803  17.6889
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -6.8477787  3.2189771  -2.127   0.0359 *
## education    4.1374444  0.3489120  11.858 < 2e-16 ***
## income       0.0013612  0.0002242   6.071 2.36e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.81 on 99 degrees of freedom
## Multiple R-squared:  0.798, Adjusted R-squared:  0.7939
## F-statistic: 195.6 on 2 and 99 DF, p-value: < 2.2e-16
```

$$\text{Errors} = y - X\beta.$$

```
err<-y-xmat%*%betas  
summary(err)
```

```
##           V1  
##  Min.      :-19.40399  
##  1st Qu.: -5.33079  
##  Median :  0.01542  
##  Mean     :  0.00000  
##  3rd Qu.:  4.98027  
##  Max.     : 17.68893
```

Note errors

```
summary(lm(prestige~education+income, data=Prestige))
```

```
##
## Call:
## lm(formula = prestige ~ education + income, data = Prestige)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.4040  -5.3308   0.0154   4.9803  17.6889
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -6.8477787  3.2189771  -2.127   0.0359 *
## education    4.1374444  0.3489120  11.858 < 2e-16 ***
## income       0.0013612  0.0002242   6.071 2.36e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.81 on 99 degrees of freedom
## Multiple R-squared:  0.798, Adjusted R-squared:  0.7939
## F-statistic: 195.6 on 2 and 99 DF, p-value: < 2.2e-16
```

Residual standard error: $\sqrt{e'e/(n-k)}$

```
ete<-t(err)%*%err/(length(err)-ncol(xmat))  
sqrt(ete)
```

```
##           [,1]  
## [1,] 7.810153
```

Note Residual standard error

```
summary(lm(prestige~education+income, data=Prestige))

##
## Call:
## lm(formula = prestige ~ education + income, data = Prestige)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.4040  -5.3308   0.0154   4.9803  17.6889
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -6.8477787  3.2189771  -2.127   0.0359 *
## education    4.1374444  0.3489120  11.858 < 2e-16 ***
## income       0.0013612  0.0002242   6.071 2.36e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.81 on 99 degrees of freedom
## Multiple R-squared:  0.798, Adjusted R-squared:  0.7939
## F-statistic: 195.6 on 2 and 99 DF,  p-value: < 2.2e-16
```

$$R^2: 1 - \text{sum}((y - xb)^2) / \text{sum}((y - ybar)^2)$$

```
1-sum(err^2)/sum((y-mean(y))^2)
```

```
## [1] 0.7980008
```

Variance Covariance Matrix Betas

```
(varcovar<-as.numeric(ete)*solve(t(xmat)%*%xmat))
```

```
##                education          income
##          10.3618135462 -1.000086e+00  1.434527e-04
## education -1.0000863848  1.217396e-01 -4.518427e-05
## income    0.0001434527 -4.518427e-05  5.027105e-08
```

Standard Errors of Betas

```
sigmas<-diag(varcovar)  
(stderrors<-sqrt(sigmas))
```

```
##                education          income  
## 3.2189770963 0.3489120233 0.0002242121
```

Note Standard Errors

```
summary(lm(prestige~education+income, data=Prestige))
```

```
##
## Call:
## lm(formula = prestige ~ education + income, data = Prestige)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.4040  -5.3308   0.0154   4.9803  17.6889
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -6.8477787  3.2189771  -2.127   0.0359 *
## education    4.1374444  0.3489120  11.858 < 2e-16 ***
## income       0.0013612  0.0002242   6.071 2.36e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.81 on 99 degrees of freedom
## Multiple R-squared:  0.798, Adjusted R-squared:  0.7939
## F-statistic: 195.6 on 2 and 99 DF, p-value: < 2.2e-16
```

tstats of Betas

```
(tstats<-betas/stderrors)
```

```
##           [,1]  
##      -2.127315  
## education 11.858131  
## income    6.070886
```

P-values of Betas

```
(1-pt( abs(tstats), df= length(err)-3))*2
```

```
##                [,1]  
##                3.587840e-02  
## education 0.000000e+00  
## income    2.355196e-08
```